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Stat 454 HW8

Ch 8; #6 (all but 1, 2b, 2g)

Do hand computation

3) Check in SAS Enterprise

Use computer to complete part a),
and compare result w/ part a),

#6.

	Sample								
Device	1	2	3	4	5	6	$\bar{y}$	s^2	CV
A	-0.357	-0.294	0.179	0.19	-0.136	-0.324	-0.0135	0.192653	-32.572
B	-0.176	0.125	-0.013	0.082	0.091	0.159	0.0446	0.043737	2.2107
C	0.137	-0.063	0.246	-0.050	0.316	0.154	0.1226	0.023479	1.2498
D	-0.062	0.640	0.21	0.166	0.219	0.407	0.2735	0.062124	0.9113

a) Both group B and D appear to have relative outliers (0.499 and 0.640 respectively).
their means are actually going to be higher than groups A and C because
they have much less negative values

b) run ANOVA to confirm/reject all conclusions. use $\alpha = 0.05$

$H_0: \mu_A = \mu_B = \mu_C = \mu_D$ H_a : at least one inequality

total sample mean = 0.114333 total sample variance = 0.060463

$$SSB = \sum_{i=1}^4 n_i (\bar{y}_i - \bar{y})^2 = 6(-0.0135 - 0.114333)^2 + 6(0.0446 - 0.114333)^2 + 6(0.1226 - 0.114333)^2 + 6(0.2735 - 0.114333)^2$$

$$= 0.252414$$

$$SSW = \sum_{i=1}^4 (n_i - 1) s_i^2 = (5)(0.192653) + 5(0.043737) + 5(0.023479) + 5(0.062124) = 1.609965$$

$$MSB = SSB / (t-1) = \frac{0.252414}{3} = 0.084138 \quad MSW = SSW / (N-t) = \frac{1.609965}{20} = 0.080498$$

$$F = \frac{MSB}{MSW} = 1.0452 \quad \text{Critical value: } F_{\alpha, t-1, N-t} = F_{0.05, 3, 20} = 3.1$$

$F < CV$, not in the rejection region so we fail to reject.
The null hypothesis at $\alpha = 0.05$, the sufficient evidence to suggest that
the population means are significantly different (seems miraculously high)

c) p-value is greater than 0.25, because the smallest value in the table w/ $\alpha_1 = 3$ & $\alpha_2 = 20$
is 1.48, which corresponds to $\alpha = 0.25$, which is greater than our chosen α , suggesting the
result of the test from SAS: p-value = 0.3466

c) the necessary conditions are:

- Each population has normal distribution: arguable, there may be a few outliers
- All population variances are the same, which does not appear to be the case based on the sample variances because they are not too close.
- Sets of observations are independent, must limit birds because there were 4 separate devices used, and the samples were randomly assigned

26. Apply Kruskal-Wallis test to PG to determine if there is a difference in the distributions of opinions across the geographical regions
 H_0 : 4 identical distributions H_1 : at least one distribution is different

$$\text{Test statistic: } H = \frac{12}{n_T(n_T+1)} \sum_{i=1}^k \frac{T_i^2}{n_i} - 3(n_T+1)$$

$$= \frac{12}{24(25)} \left(\frac{T_1^2}{9} + \frac{T_2^2}{5} + \frac{T_3^2}{7} + \frac{T_4^2}{9} \right) - 3(24+1)$$

$$= \frac{12}{24(25)} \left(\frac{9^2}{9} + \frac{6^2}{5} + \frac{7^2}{7} + \frac{10^2}{9} \right) - 75$$

$$= 20.48$$

Rank Table

1256											Rank				T:					
A	2	3	24	11	5	1	5													
B	4	12	9	10	11	22	68													
C	13	6	14	7	20	14	79													
D	8	12	17	15	18	21	102													

Critical value: $\chi^2_{0.05, 3} = 7.815$

Rejection region: $H > 7.815$



$20.48 > 7.815$, so we reject

the null hypothesis at $\alpha = 0.05$,

meaning at least one of the distributions or all four is different, and must have different population variances, or means

Power for F test

→ significant results suggests the data is not skewed or heavily tailed. p-value of the Kruskal-Wallis test is different to the value calculated in 8c

	$\bar{X}$	S^2
herbicide 1	89.22	71.449
herbicide 2	90.246	26.176
Control	87.423	54.9377

a) $H_0: \mu_1 = \mu_2 = \mu_3$ $H_1: \text{At least one}$ $\alpha = 0.05$

total sample mean = 87.64999 total sample variance = 57.7378

$$SSB = 30(89.22 - 87.64999)^2 + 30(90.246 - 87.64999)^2 + 30(87.423 - 87.64999)^2 = 2364.49$$

$$\rightarrow MSB = \frac{2364.49}{2} = 1182.245$$

$$SSW = 20(71.449) + 20(26.176) + 20(54.9377) = 4409.8157$$

$$\rightarrow MSW = \frac{4409.8157}{67} = 65.8187$$

critical value: $F_{0.05, 2, 67}$
 $= 3.101$

$$F = \frac{MSB}{MSW} = \frac{1182.245}{65.8187} = 17.96$$

$17.96 > 3.101$, so we reject the null hypothesis

at $\alpha = 0.05$, suggests that (at least one inequality) between μ

5) 99% CIs in each mean

herbicide 1: (82.2512, 88.3007)
 herbicide 2: (88.4151, 92.0769)
 control: (84.7827, 90.0632)

$$\bar{X} \pm Z_{\alpha/2} \frac{S}{\sqrt{n}}$$

$$Z_{\alpha/2} \rightarrow Z_{0.005} = 2.58$$

c) ~~herbicide 1~~ both appear to be different from the control, but both herbicides overlap the control at the control group, herbicide on the lower side, herbicide 2 on the upper side

I'm not sure if I did any of these problems correctly but I also know where I may have gone wrong